MassMarsh
Standard Operating Procedure:
Collection of Hydrologic Data

1. Scope and Application

A standard set of procedures is established to set up a monitoring effort to assess tidal hydrology, the primary driver of salt marsh development and health. Cape Cod divides Massachusetts into mesotidal (2-4 m tidal ranges) for the outer cape and northward and microtidal (< 2 m tidal ranges) for Vineyard Sound and marshes to the southwest. Since European settlement 400 years ago, salt marsh tidal hydrology has been altered in two dramatic ways. First, barriers have been placed across marshes (transportation corridors, agricultural embankments, dikes, etc. with hydrologic controls), reducing tidal exchange, including sediments and salinity (Adamowicz et al. 2019). Second, marshes have been ditched for a variety of reasons, resulting mostly in increased marsh porewater drainage at low tide, though in some cases reducing surface water drainage when ditch spoils form circumferential low dikes around marsh units. Many of the restoration approaches commonly used today have a focus on providing more natural hydrology to the marsh. Therefore, any assessment of marsh health and condition needs hydrologic data to contextualize vegetative condition and response to restoration. Procedures detailed in this document can be applied to monitoring surface water or groundwater elevation.

Impacts to vegetation occur from tidal flooding but also from the depth and duration of peat saturation as it affects plant roots, which are dependent on oxygen for respiration (Morris et al. 2002). Automated water level recorders (WLRs) are typically set up to record data 4 - 12 times an hour, defining a 12.5-hour sinusoidal curve that shows the rise and fall of tides in an unrestricted tidal creek. Smaller creeks and ditches farther up into the marsh as well as restricted tides exhibit asymmetrical tidal curves, with ebbing tides requiring more time as the water slowly drains from the marsh. WLRs set up to measure the frequency and duration of flooding at a specific point in the marsh are installed in the peat, though water levels in the housing may not accurately reflect the water levels in the peat due to low hydraulic conductivity of the peat (more experimental work is required). Despite this uncertainty, groundwater WLRs can yield several measures of flooding that affect the vegetation. Because each species has evolved their own adaptations and abilities to thrive in tidal marshes, variation in flooding and salinity stressors across each marsh have led to the development of recognizable marsh zones characterized by the dominant species.

2. Summary and Approach

When installed in a marsh creek or groundwater well for more than a month, WLRs provide an inexpensive means to characterize the tidal hydrology within a marsh. The data produced can be statistically reduced to yield an overall frequency and duration of flooding broadly and at specified surveyed locations within a marsh as part of a comprehensive wetlands assessment. Restoration of hydrology can come in many forms, dependent on the downstream or within-marsh impairment.

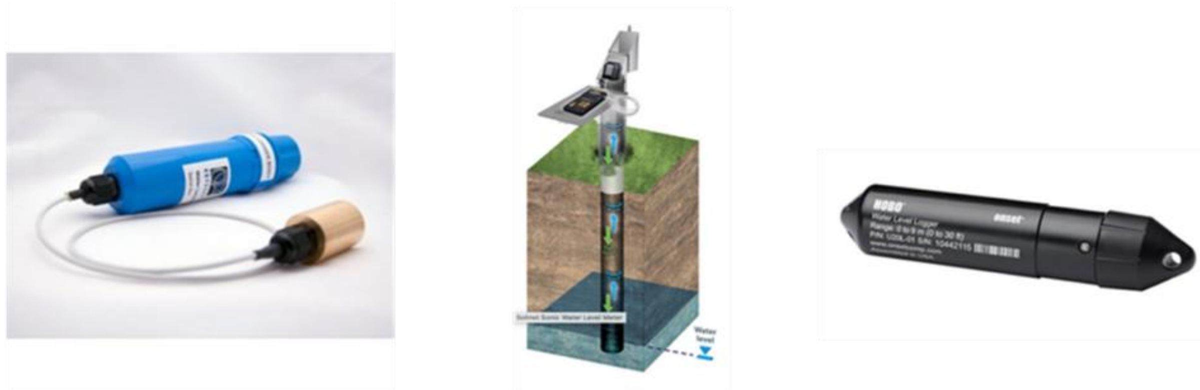
Instruments on either side of a tidal restriction can be used to calculate hydraulic head differentials and estimate tidal asynchrony. Combined with geomorphic data, tidal exchange can be modeled to propose a range of dimensions of the crossing structure to help select an appropriate design (Moore et al. 2023). Following tidal restoration, WLRs can be set up in a similar fashion to assess whether restoration achieved the hydrologic goals.

In cases of restoration actions including sediment placement and/or improvement to surface hydrology using runnels and ditch remediation, WLRs can be installed to capture changes resulting from restoration actions. WLR observations can then be used to assess performance and inform adjustments to the restoration actions (e.g., “adaptive management”).

3. Materials and Equipment

Before leaving for the field the Field Manager will confirm the following equipment is available, charged and in operating condition:

A. Real-Time Kinematic GPS and related accessories to establish instrument elevation to 2-3 cm vertical accuracy as described in Appendix A-6C.
B. Extra batteries for antennae and GPS
C. Water level recorders (Onset, Solinst, or similar) pre-programmed to start recording prior to field deployment
D. WLR mounted in 1 ¼ inch pvc for within peat measures
E. Post with WLR enclosure (stilling well) and attachment hardware for tidal creek measures
F. Mobile Phone (time, flashlight, camera, google maps)
G. Datasheets and clipboard, including field notebook
H. Tool to measure water depth over WLR while in situ (e.g., meterstick with tubing, measuring tape with water finding paste)
I. Appropriate clothing, food, and water during field work (see safety list)
J. Cable ties
K. Pipe of pvc, 1 ¼ inch diameter in 2-3’ sections to mark well locations after instruments are removed
L.



Capacitance-based Acoustic Pressure

Figure 1. Types of water level recorders useful for measuring tides.

4. Training and Calibration

All field crew leaders will be scientists with experience conducting research in salt marshes and will be familiar with safe installation of WLRs in waterways and groundwater wells. Field technicians will be trained and supervised by crew leaders.

All equipment will be calibrated and maintained according to manufacturer recommendations. Based on the equipment (capacitance, acoustic, or pressure transducers), it is always recommended each piece of equipment is evaluated for calibration, internal memory storage, and battery life before the upcoming field season. For example, pressure transducers can be tested by placing them in known depths of water that are varied over several hours and then the data plotted to assess calibration. It is recommended that monitoring teams maintain a spreadsheet on the status and deployment history of their WLRs. The spreadsheet ensures that uncalibrated or non-functioning WLRs are not mistakenly deployed in the field and notes when equipment breaks or malfunctions in the field.

Before deployment, scientists should be aware of the storage capabilities and battery life of their equipment in relation to deployment and retrieval dates as well as the data collection frequency. Increased monitoring frequency (e.g., 10 min to 5 min intervals) will decrease duration prior to the logger reaching its data capacity. Logger deployment software typically provides maximum duration length when preparing the equipment for deployment.

Pressure transducer WLRs that are not attached to a vented cable require a barometric adjustment for atmospheric pressure. Typically, a pressure transducer is placed in a nearby tree (within 4-6 kilometers) and is used to correct the WLRs for variation in atmospheric pressure. Hydrology monitoring equipment is fragile and sensitive to shock; the equipment should be stored safely and handled with care.

5. Procedures

5.0 Selection of measurement sites

5.01 Monitoring to determine marsh condition

Using aerial photographs to examine the project area, select an appropriate location to show surface water elevations within a major tidal creek and at the requisite number of locations to measure groundwater levels within the peat (Figure 2). To get an accurate sense of the timing and magnitude of tidal stage changes, measures should be taken to install a water level recorder as low in the tidal frame as is feasible and safe. The tidal creek should be the primary hydrologic pathway that drives tidal exchange. The creek WLR should be installed about a third of the distance up the creek into the marsh and as close to the creek thalweg (deepest location) as possible. Safe installation in large creek systems may require deployment from the marsh bank, therefore reaching the thalweg is preferred but not necessary, especially since the low tide details are not of critical importance unless monitoring for tidal flow restoration (discussed in detail below). To assess porewater elevations in the marsh proper, at a minimum, plan to establish a well in the peat of an area that appears to be healthy and another at a site that appears to show some type of impairment.

It is of great value if several different types of data can be interpreted together (e.g., changes in ground water affecting plant community composition). For example, if vegetation and elevation transect surveys are conducted for a given area, it is recommended that groundwater wells be placed in areas between the transects (Figure 5). Transect elevations can then be used to calculate the marsh platform elevation for hydrology data analysis, and the vegetation data can then be viewed and understood through the lens of surface and groundwater hydrology.

5.02 Monitoring to determine effectiveness of marsh restoration

Restoration of tidal exchange

Using the tidal flow restoration design plan, identify appropriate locations for surface water WLRs both upstream and downstream of the tidal restriction (Figure 3). WLR elevations should be as low as is feasible and safe to effectively capture the entire tidal range immediately upstream and downstream of the restriction. Ensure the WLRs will not be disturbed or damaged based on the construction access, staging, or restoration plans. In addition, install a WLR in a minor or secondary ditch or creek upstream in the marsh to assess the impacts of restoration on daily high tide flood timing and elevations. If there are areas of concern or interest on the marsh platform (e.g., waterlogged vegetated pannes, pools, etc.), wells can be established in the peat to monitor the impacts of the tidal restoration on groundwater hydrology.

Sediment deposition to increase elevation

Sediment placement is being tested as a restoration method to increase the elevation of marshes thought to be in danger of drowning. Knowledge about changes in hydrology following sediment placement is severely lacking and hydrology has been suggested as a reason for poor revegetation and vegetation loss after such interventions. Surface water elevations are needed from a tidal creek seaward of the placement to determine the flooding and draining potential. Several wells placed in peat could be aligned in a transect from creek to areas distant from any obvious drainage paths. Such data could help illuminate the drainage requirements of high marsh plants following sediment placement.

Restoration of surface hydrology using ditch remediation, drainage enhancement and habitat islands

From the restoration design plan for the marsh, choose locations for WLRs in the creek and marsh platform for groundwater to reflect the marsh response following management actions. A surface water WLR is advised for the major tidal creek to understand the tidal regime at the site and provide context for measurements of surface and peat hydrology (Figure 4). If parallel ditches are to be remediated, a well could be installed in the central marsh platform between ditches less than 12 m apart to assess the effect of the remediation. Data collected from year to year should show increases in the groundwater table but not increases high enough to kill the plants. If a runnel is created to drain a panne area, a groundwater well could be placed nearby in the peat to determine how much the water table is brought down by the drainage and if the tidal cycle is more supportive of high marsh grasses. Establishment of WLRs in the peat will help illuminate the drainage requirements of high marsh plants.

Figure 2. Example of a hydrology monitoring design to explore pre-restoration conditions at Little River Marsh, NH. Groundwater wells were placed at several waterlogged or pool locations to evaluate degraded conditions and two reference high marsh platforms to represent goals for restoration. Two creek WLRs were deployed to assess site-level tidal hydrology.

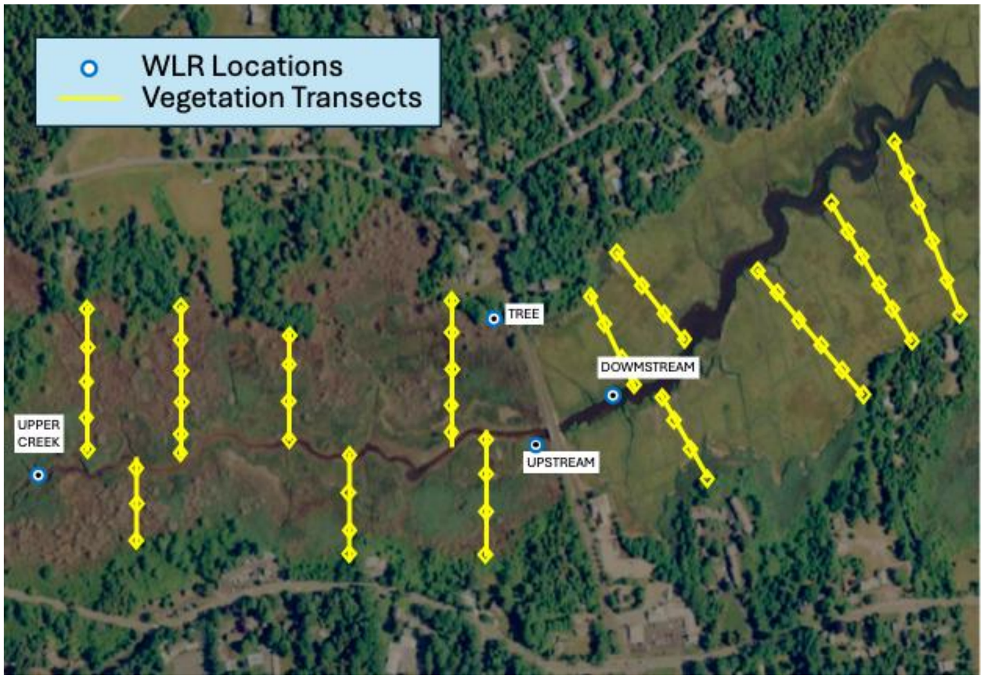


Figure 3. Monitoring design example for WLR locations to gauge before and after culvert expansion (Upper Creek, Upstream, Downstream, Tree) at Sesuit Creek, Massachusetts.

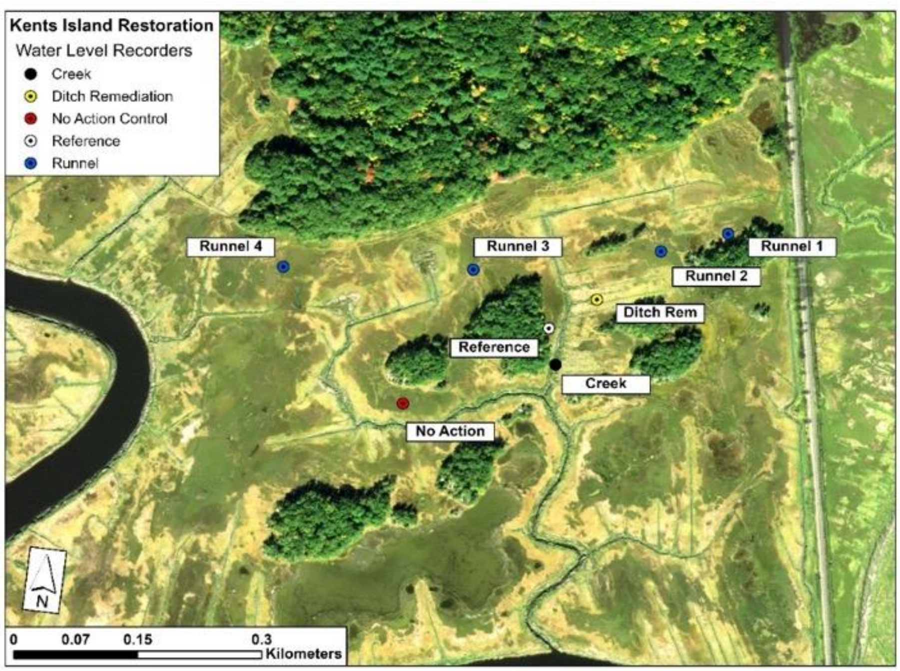


Figure 4. Example of a monitoring design at marsh south of Kents Island to evaluate the effectiveness of surface hydrology restoration with runnels and ditch remediation compared to reference high marsh and no action control treatments. A WLR was placed in a creek central to the study area to assess site-wide tidal hydrology.

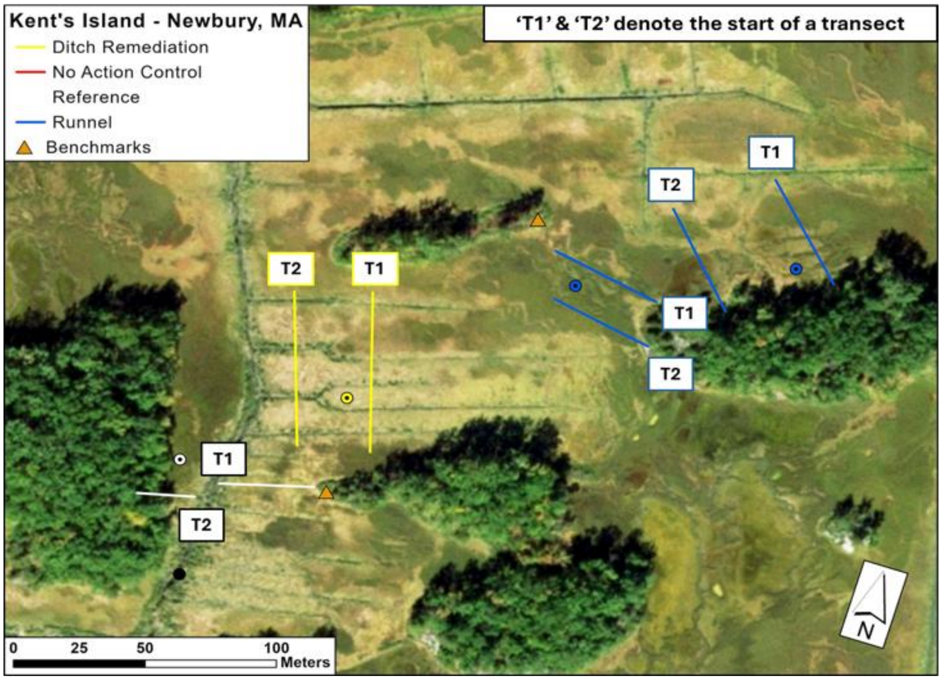


Figure 5. Example of hydrology monitoring design concurrent with vegetation and elevation transects at Kents Island Marsh, MA. Where possible, groundwater wells (circles) were located between the transects of each treatment.

5.1 Monitoring Timeframe

Most monitoring protocols have recommended a minimum period of one lunar cycle (29.6 days) for measuring tidal hydrology using WLRs (Neckles and Dionne 2000, Moore et al. 2023). Based on five years of monitoring hydrology, it is not clear whether one complete lunar cycle is adequate to accurately gauge the tidal hydrology of a given site, especially for groundwater hydrology. Seasonal (droughts, excessive rainfall), annual (several large storms) and long-term variation (18.6-year metonic cycle) can create enough noise to drown out subtle long-term signals within short timeframes. Therefore, hydrologic changes from restoration actions can be difficult to discern if data are collected from only one lunar cycle. If equipment availability and funding allow, it is recommended that scientists deploy WLRs for as long as the growing season (May – September) to capture the full hydrologic conditions experienced by the vegetation community. If this is not possible, 2 – 3 tidal cycles (60 – 90 days) may be sufficient to describe the tidal conditions in tidal creeks and peat groundwater with the understanding that climactic variations may impact results.

5.2 Water Logger Installation

Hydrological records for creeks, ditches, and pools are obtained by placing the WLR in a protective casing (e.g., pvc tube with plenty of holes) and fixing the case to a post with stainless steel pipe clamps (Figure 6). Protective cases are recommended to protect equipment from floating debris, ice rafting, and wildlife during deployment. Cable ties can be used, but they can become brittle and break in exposed locations. The case should be set so that the WLR is as low as possible without becoming clogged with mud, about 2 cm above the bottom. Because pools can dry out (interrupting the record), place in the deepest portion of the pool and/or push away the mud from the case. Elevations are determined by placing the base of an RTK or measuring rod on top of the case and knowing the WLR elevation relative to the top of the case (see more details in 5.3 Elevation Surveys).

Monitoring groundwater elevations (i.e., the watertable) in salt marshes involves installing a perforated pipe into the marsh platform and recording continuous water levels in the pipe (or well bore) with a WLR. Low hydraulic conductivities in marsh sediments (especially muddy marshes) coupled with rapid changes in tidal stage make it challenging to accurately observe the height of the water table in salt marshes (Gardner, 2009). Specifically, the rate at which water flows into or out of a well bore can be limiting, such that the water level inside the well differs from the true water table depth outside the well bore. The following recommendations from Gardner (2009) can minimize the difference between water table elevation and that recorded inside your well bore:

- Minimize the well bore inside diameter, making it as small as possible to accommodate your pressure transducer. This minimizes the volume of water that needs to pass through the well screen (perforated section);
- Maximize the length of the well screen (60-90 cm is appropriate);
- Excavate a hole at least double the radius of the outside of the well using an augur;
- Pack sand around the well bore that fills the hole's annulus, tamp with broom stick;
- Seal the sand pack with native marsh material, at least 15 cm (6 in);
- Be sure that the screened interval does NOT extend above the marsh platform;
- The well stickup should extend above MHHW to assure water does not enter the well bore from above; if impractical (e.g., low marsh deployments), drill a 1-2 mm hole to prevent a vacuum, while minimizing water entry when submerged;
- Tie the WLR to a string or cable and affix the string to the top of the well. Drilling a 2 mm hole at the top of the well can provide a simple tie point;
- The WLR can be suspended just above the bottom of the well (non-stretch string), or rest on the well bottom with the transducer vent oriented upwards to prevent clogging.

Surveying water elevation should happen at least 30 minutes after the well installation to allow the water level to stabilize somewhat. Measure the depth to water below the well stickup and subtract the elevation of the well stickup above the marsh platform. Measuring the distance from the stickup to the water surface is most easily and quickly done by blowing air through a piece of flexible tubing attached to a meter stick that is inserted into the well. Upon hearing bubbles, carefully find the minimum insertion distance that causes bubbles and record that distance. Water finding paste on a graduated dip stick (ruler) can also be used. Be sure to make measurements synchronous with a WLR observation (observations on 10 min intervals facilitate this). In low hydraulic conductivity muddy sediments, equilibration of the well water level with the water table can take several hours to days. For this reason, initial observations may be used for calibration but should not be considered when analyzing water table trends.

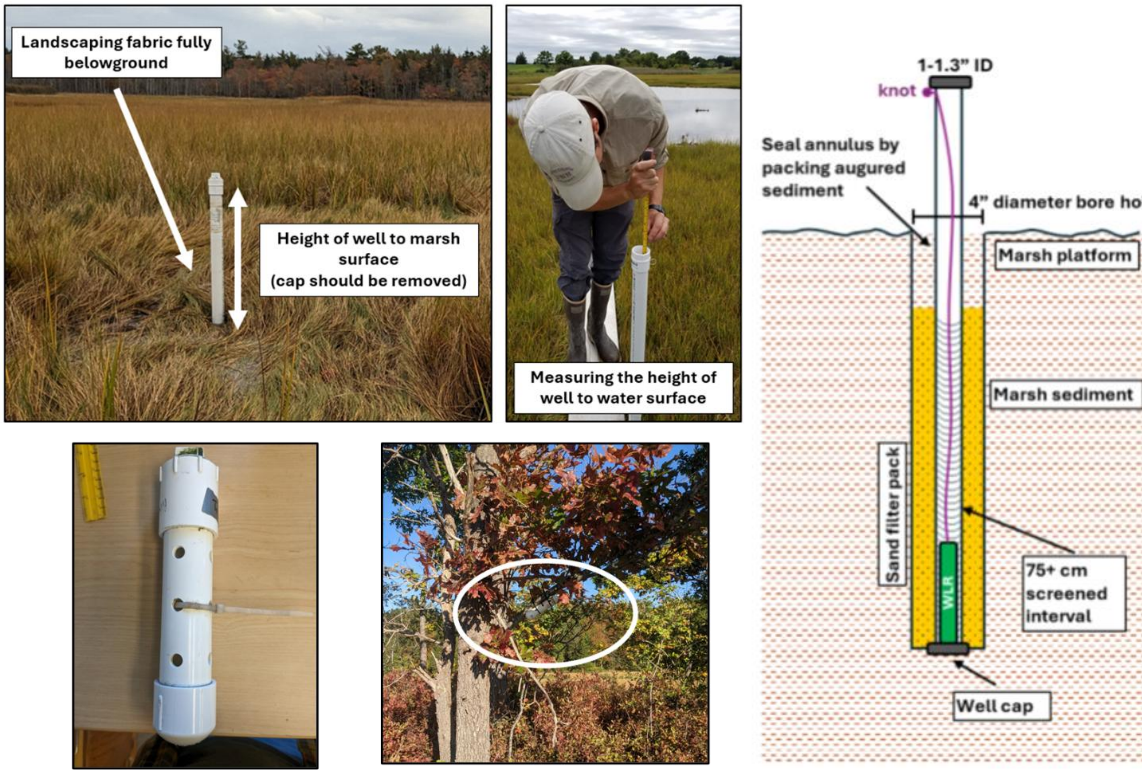


Figure 6. **(Top Row)** Example of a groundwater well installed into the marsh platform with a pressure transducer placed in the bottom of the well. The height of the rim above the marsh surface (left) and water surface within the well (center) can be measured using different techniques. **(Bottom Left)** Example of a PVC protective case for pressure transducers deployed in creeks, pools, and trees for background barometric pressure compensation. **(Bottom Center)** Deployment of a pressure transducer for barometric pressure on a tree in adjacent uplands. **(Right)** Example of well design parameters, including screened interval, sand filter pack, annulus diameter, and sealed annulus. A WLR is resting on the well bottom with vent oriented up to prevent clogging (MassMarsh 2025).

5.3 Elevation surveys

Elevations surveys will be taken to (1) measure the elevation of the marsh platform surrounding WLRs and (2) survey water level observations to a global datum (*i.e.,* NAVD88 m). Elevation surveys can be conducted with a real-time kinematic GPS (RTK-GPS), laser level, or theodolite that is then tied into a true elevation benchmark such as USGS or state geodetic benchmarks (See Sediment – Elevation SOP). We do not recommend using LiDAR as a substitute for on-the-ground field surveys for elevation due to the vertical accuracy error associated with salt marsh vegetation cover in most LiDAR products. The elevation of the marsh platform is crucial for calculating flooding and drainage statistics (see Section 5.4). The elevation of the marsh platform can be derived from several methods. First, the elevations from vegetation plot monitoring of transects within a given area can be obtained and averaged using a laser level or theodolite (tied to a benchmark). The placement of a well between two transects of a given area (see Figure 5) will ensure the transect elevations and groundwater hydrology will generally agree. Second, for groundwater wells or specific locations of interest, a 'spot' elevation can be taken, and the single value can be used for flooding statistics.

Intertidal WLR’s can be installed subaerially or sub-aqueously depending on tidal timing and position within the tidal frame. In many cases, subaerial installation at low tide will be safest and most straightforward. Read the manual for your WLR to determine the precise location of the pressure sensor vent and survey in the elevation of the vent after affixing the WLR to a post. If deploying the WLR sub-aqueously in a well or channel, you will need to survey in the elevation of the water surface synchronously with a WLR recording as tidal stage can change quickly. This elevation should be recorded on the day of installation and the day of instrument recovery (Appendix 1).

5.4 Data Processing and Analysis

Data Processing and QA/QC

Common software for equipment can transform capacitance, pressure, or acoustic data into water elevations (meters), not tied to a specific vertical datum. Scientists can then survey water surface elevations to a global vertical datum such as the North American Vertical Datum 1988 meters (NAVD88 m) based on reference water elevations measured upon deployment and collection of instruments. Software like Hoboware™ provides that function during data processing. Transforming water elevations into true elevation vertical datums is critical for multi-year or multi-site comparisons as well as disseminating the data to stakeholders, colleagues, and the public.

QA/QC protocols have been developed to ensure accurate comparisons of water level profiles between sites and across field seasons. When multiple WLRs are deployed across a site, it can be assumed that all WLRs will be flooded to the same elevation for the highest spring tide. After data processing, the water elevation profiles of any deviant WLRs can be uniformly shifted by the difference in spring tide elevations of it and the primary creek WLR. The spring tide shift, expected to be less than +/- 5 cm, compensates for the error associated with RTK-GPS or laser level.

Data Analysis

Water elevations can be graphed and analyzed using different software including Microsoft Excel, R, Matlab, and Python. software. It should be noted that several packages in R have been specifically developed to analyze tidal hydrology of creeks and ditches and calculate common tidal datums (MLLW, MLW, MTL, MHW, MHHW, and highest tide elevation; Figure 7) such as *VulnToolkit* (Hill and Anisfeld, 2015) and *Tides* packages (Cox and Shepherds, 2018). Based on elevation surveys (Section 5.3), *VulnToolkit* can calculate the high tide flooding frequency and flooding duration for marsh platform and root zone from both surface water and groundwater WLRs (Table 1, Figure 8). The *VulnToolkit* has additional capabilities of retrieving and downloading historic tidal data from NOAA tidal gauges. Several flooding and drainage parameters have been developed for comparisons across treatments and over time:

- *Flooding Duration (days or %)* – the percentage of time the marsh platform or root zone elevation is submerged for the monitoring period.
- *High Tide Flooding Frequency (%)* – the percentage of high tides (based on either nearby creek or groundwater wells) that inundates the marsh platform or root zone elevation.

- *Drainage depth (m)* – the mean depth groundwater drains for each tidal cycle relative to the marsh platform or root zone. Drainage depth is calculated as the mean low tide elevation subtracted from the marsh platform elevation. Positive values indicate, on average, groundwater does not drain below the marsh platform.
- *Drainage amplitude (m)* – an elevation-blind metric that describes the change in the shape of the water elevation profile. The metric may be useful for comparison of pre- and post-restoration of pools and shallow water pannes, where little variability is seen with water elevations (*i.e.,* constant water elevation). Drainage amplitude is calculated as the mean low tide subtracted from the lowest water elevation observed in monitoring.
- *Median Duration of Flooding Events (hrs)* – the mean duration of time water levels falls below the marsh platform or root zone elevation after each flooding event. The measurement illustrates how quickly surface or groundwater drains from the local area. It is recommended that this metric be calculated with water level deployments for 2 – 3 months to accurately capture multiple spring and neap tidal cycles.

One major consideration for data analysis of groundwater wells is that *Tides* and *VulnToolkit* packages struggle to identify timing and elevations of low tides in groundwater wells due to the asymptotic pattern associated with drainage that occur either continuously until the next flooding event or not at all in impaired systems. Scientists will need to identify the "low tide" for each tidal cycle. The low tide can be calculated based on timing of the low tide with nearby creek WLRs or classified as the lowest water elevation between two high tide events.

Other metrics can be developed from the hydrologic data such as the longest number of days or hours a specific elevation is continuously flooded in the marsh (Neckles et al. 2013), or the time elapsed between flooding events for a specific elevation. For example, length of time between floodings for ground nesting sparrows can be estimated by comparing the average elevation of a particular marsh area with the water level profile (Figure 9). Twelve days elapsed between flooding on 9/25 and 10/8, barely enough time for sparrows to lay eggs, incubate until hatching, and be able to survive out of the nest.

Table 1: Flooding statistics for individual groundwater water level recorders at Old Town Hill Marsh 2021 – 2024. High tide flood frequency is the percentage of high tides that reach the marsh platform or root zone elevation in each monitoring period (of a total of 56 high tides). Flood duration is the percent of total monitoring time the marsh platform or root zone elevation was flooded (of a total of 710 hours).

		Marsh Platform		Root Zone	
Groundwater WLR	**Year**	**High Tide Flood Frequency (%)**	**Flood Duration (%)**	**High Tide Flood Frequency (%)**	**Flood Duration (%)**
Reference 1	2021	12.7	67.2	45.9	84.5
(SC3)	2022	29.4	66.7	62.1	86.0
	2023	15.9	64.4	30.7	71.2
	2024	19.0	69.0	56.7	84.5

Reference 2	2021	2.3	15.5	3.8	22.4
(NNC)	2022	3.5	24.6	28.9	31.6
	2023	5.4	31.7	55.2	12.1
	2024	1.8	8.5	6.9	16.9

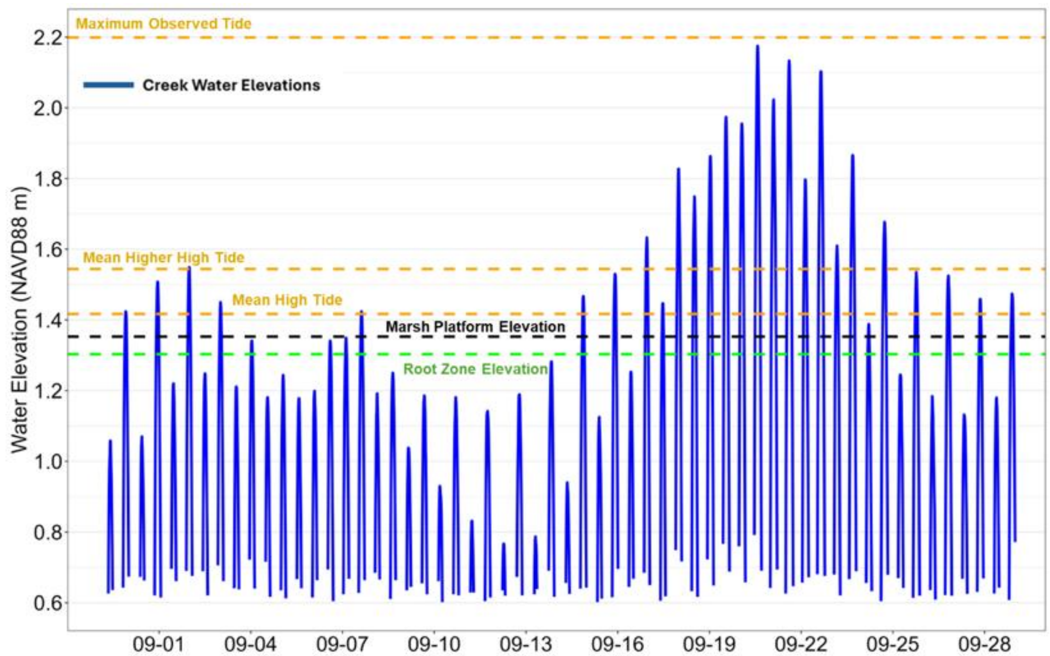


Figure 7. Visual description of commonly calculated tidal datums including mean high tide, mean higher high tide, and maximum observable tide (orange dashed lines) based on water elevation profile (blue line) from a creek in John Wise Marsh (Essex, MA) in 2024.

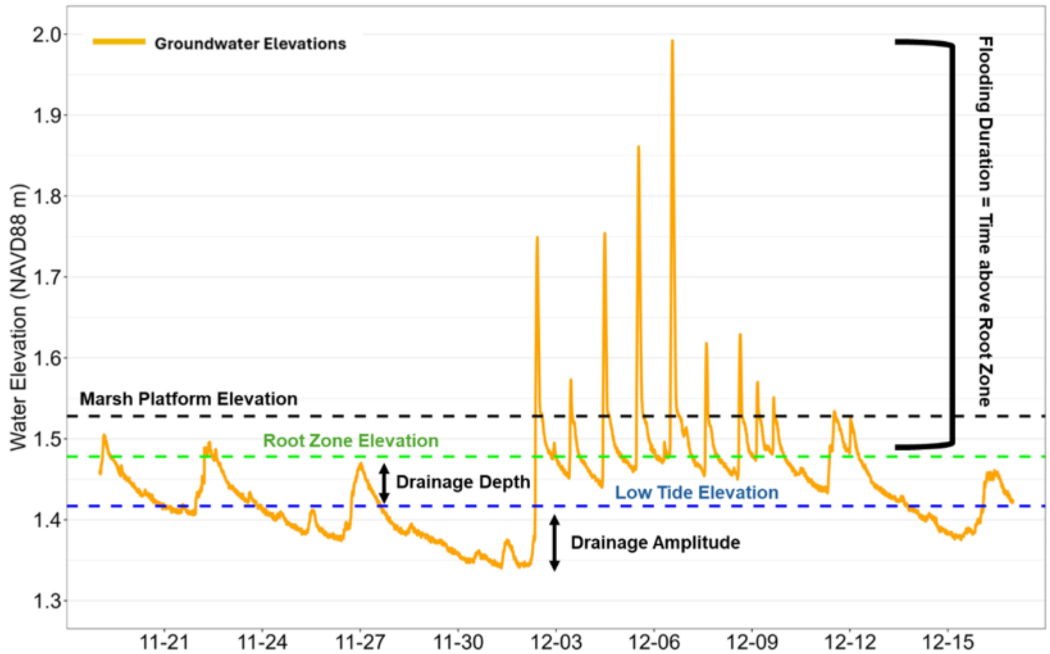


Figure 8. Visual description of the recommended hydrology metrics including flooding duration, drainage depth, and drainage amplitude calculated for the root zone (5 cm below the marsh platform). Groundwater profile for a runnel restoration area in Kents Island Marsh (Newbury, MA) is provided in orange. The low tide elevation is calculated as the average minimum water elevation between each high tide during monitoring.

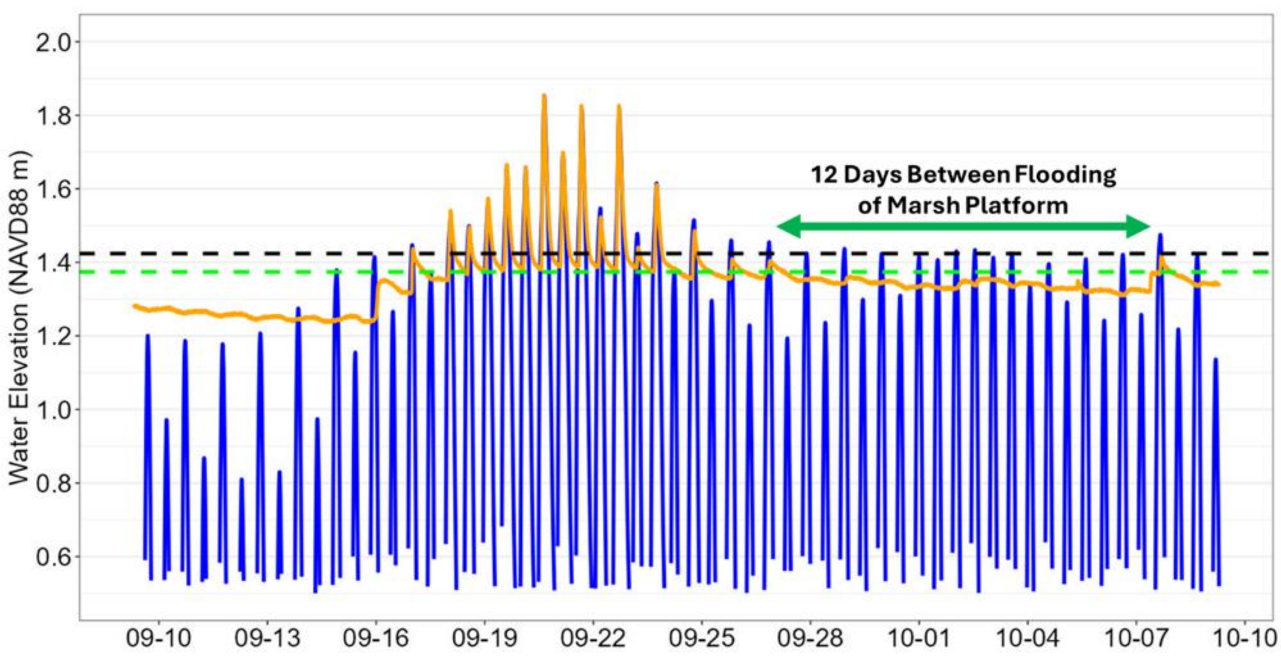


Figure 9. Example of a water elevation profile at a runnel area for one tidal cycle in Fall 2024 at Kents Island Marsh. Reference elevations are provided for the marsh platform (black dashed) and root zone 5 cm belowground (green dashed).

5.5 Protocol for Decontamination of Field Equipment

All equipment will be inspected for debris before leaving a site. Debris removed from boots and equipment will be disposed in a trash bag or on dry, high ground. When possible, equipment will be left to air dry and inspected to remove any remaining plant fragments. Waders will be sprayed with 10% bleach solution, scrubbed, and rinsed with tap water to remove any additional debris.

6. Quality Assurance/Quality Control

All field crews will be led by field personnel experienced in salt marsh field research and plant identification. Other technicians will be trained to fulfill responsibilities in the field and adhere to QA/QC procedures. Two field crews will initially work together to collect field data until consistent results are achieved between the two teams, after which the teams will work independently.

7. Interferences

Inclement weather (heavy rain and strong winds) may interfere with our ability to collect representative data on a variety of parameters. Severe weather may delay field data collection due to safety concerns. Access may be a challenging aspect of data collection, especially in more developed areas of the study area.

8. Trouble Shooting

Using a recording interval of 10 minutes and a start time on the hour allows the field team to always know when they should take calibration measurements in the field. Be sure to set the barometric pressure recorder to the same record interval and start time to allow for simple barometric compensation calculations.

9. References

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10. Safety Considerations

- Fieldwork will not be conducted during heavy rain events or unsafe conditions such as electrical storms or high wind events.
- Field personnel will not access sites alone without the instruction of a field manager.
- All persons must carry cell phones or other emergency communication devices while sampling. It is recommended they be waterproof or stored in a waterproof case.
- If there is no safe access to a plot point, the field sampling will either be moved to a location with safe access or not be conducted for that site if a replacement location is unavailable.
- If a boat, canoe, or kayak is used to for access, all persons must wear personal flotation devices while on the water.
- All watercraft used must comply with USCG required safety equipment per 46 U.S.C, Chapter 43. An online Vessel Safety Check interactive program is available from USCG at http:/www.uscgboating.org/safety/vsc.htm.
- Good judgment will be used in selecting clothes and personal protection items. Common items needed include: extra clothing, sunshade, sunscreen, hats, insect repellent, and waterproof knee boots, chest waders or hip waders for highest anticipated depths. Any staff not dressed appropriately for fieldwork should not participate in the survey. Proper footwear is a must (e.g., no “flip-flops” for field work).
- Good judgment will be used in walking on marsh surfaces; mosquito ditches will be circumvented or crossed with caution.
- Each crew member will carry a personal first aid kit.
- No chemicals will be handled by personnel in the field (decontamination will occur in parking areas or after returning from the field).
- Private property will be respected using the following guidelines.
    - Landowner permission to access property will be sought for all sites.
    - If property is in close proximity to buildings or other heavily used areas, the site will not be assessed without landowner permission.
    - Posted property will not be accessed without permission of the landowner.
    - If asked to leave private property by the landowner, samplers will discontinue work and leave.

11.1 Data to be Generated

Hydrology monitoring creates several datasets including field datasheets, unprocessed time series data (e.g., raw pressure, acoustic, or capacitance data), processed time series water elevations, and statistically reduced datasets for each monitoring season (e.g., tidal datums, flooding duration, etc.). Additionally, it is recommended that scientists maintain an active database on the deployment history of each WLR. Due to the volume of data inherent to hydrology monitoring, scientists should maintain an organized database (or system of folders) for each site that contains the datasets from each step of data processing and analysis. Based on prior experience, scientists should have the following folders for each site and year of monitoring:

- Field Deployment Datasheets (or same folder as vegetation datasheets)
- Deployment Data
- Unprocessed time series data
- Processed time series water elevations
- QAQC
- Hydrology Datums and Statistics

11.2 Data Handling, Storage and Backup

At the end of each day, completed data sheets will be immediately photographed by a field technician. All photographs will be and uploaded to the project folder which is stored systematically and in a manner that is continuously backed up to the cloud under the correct Site/date folder (MassMarsh Data Archive), as follows:

Root folder: Root folder:
<<SITE NAME>>
> <<YEAR>>
> FIELD MONITORING DATA
> Hydrology_SiteName_Year_Month_Day

Completed waterproof datasheets will be stored in the project filing cabinet. Images of waterproof data sheets will be stored in project Google Drive. As soon as possible, upon return to the lab or when Internet access is available, the field technician will enter all data from each waterproof plot form and the RTK unit into a Google Drive spreadsheet. Data from the plot sheets and RTK (including the spreadsheet) will be backed up using a project-issued external drive. This results in four levels of redundant plot sheet backup: (1) the waterproof copies stored in the project file cabinet; (2) images of these plot sheets stored in the Ground Truthing folder for the site day, month and year in Google Drive; (3) the data entered data in the Google spreadsheet stored in the same Google Drive folder, and (4) the data sheet photos and the spreadsheets downloaded to an external hard-drive.

Electronic hydrology datasets (see Section 11.1) will be immediately stored in their respective folders on Google Drive after creation. Data will be backed up using a project-issued external drive. The hydrology data will uploaded to the folders as follows:

<<SITE NAME>>
> <<YEAR>>
> HYDROLOGY MONITORING DATA
> UNPROCESSED TIME SERIES DATA
> PROCESSED WATER ELEVATIONS
> QAQC
> HYDROLOGY DATUMS AND STATISTICS

11.3 Project permanent archive

The project will use the Open Science Framework (http://osf.io) cloud-based system to store all final project documentation and final datasets. Final datasets include classification outputs, accuracy assessments, maps, etc. This cloud-based system will not be included as a routine data management backup folder. OSF has established a site preservation fund and advertises that the fund is sufficient to ensure the site is up and operational for 50+ years (see http://help.osf.io/m/faqs/l/726460-faqs).

Water Level Recorder Deployment and Collection Field Sheet

Site, Town, State: ____________________________________ **Date:** ________________________ **Field Monitors:** _________________________

Hobo Data Recording Interval: _____________ **Deployment or Collection:** __________________ **Other Info:** _____________________________

Water measurements are taken on 10-minute intervals (e.g., 6:00 pm, 6:10 pm, etc.) to align with Hobo pressure transducer data recording

Site and Treatment Information			Elevation and Waypoints		Time of Water Depth (Both WLRs)	Creek WLRs	Groundwater WLRs			
Area & Treatment	Type	WLR #	RTK / GPS Waypoint (Circle One)	RTK Elevation (NAVD88 m)	Time Water Elev. Taken	Water above/ below nut (m)	Well Rim to Water (m)	Well Rim to marsh (m)	Well Rim to Sensor (m)	Well Length (m)

Water Level Recorder Deployment and Collection Field Sheet

Site, Town, State: John Wise Marsh, Essex, MA **Date:** August 13, 2024 **Field Monitors:** David Burdick, Grant McKown

Hobo Data Recording Interval: 10 minutes **Deployment or Collection:** Deployment **Other Info:** ______________________________

Water measurements are taken on 10-minute intervals (e.g., 6:00 pm, 6:10 pm, etc.) to align with Hobo pressure transducer data recording

Site and Treatment Information			Elevation and Waypoints		Time of Water Depth (Both WLRs)	Creek WLRs	Groundwater WLRs			
Area & Treatment	**Type**	**Serial Number**	**RTK / GPS Waypoint (Circle One)**	**RTK Elevation (NAVD88 m)**	**Time Water Elev. Taken**	**Water above/ below nut (m)**	**Well Rim to Water (m)**	**Well Rim to marsh (m)**	**Well Rim to Sensor (m)**	**Well Length (m)**
Creek	Creek	21410806	50	0.350	8:30 AM	-0.190				
No Action	Groundwater	21182445	51	2.111	9:20 AM	N/A	0.830	0.810	1.500	1.500
Runnel	Groundwater	21410811	52	2.002	10:00 AM	N/A	0.790	0.800	1.500	1.500
Ditch Remediation	Groundwater	21182452	53	1.904	10:40 AM	N/A	0.700	0.750	1.500	1.500
Reference	Groundwater	21773462	54	2.131	11:30 AM	N/A	0.850	0.800	1.500	1.500
Tree (Barometric)	Barometric	20915045	55	N/A	N/A	N/A	N/A	N/A	N/A	N/A